

LISTA DE EXERCÍCIOS AB & CD

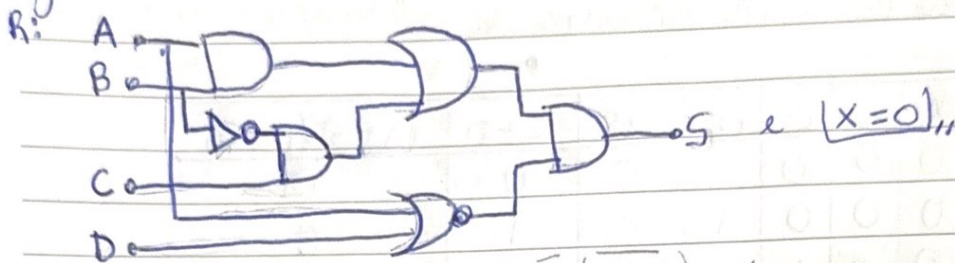
PORTAS LÓGICAS e ALGEBRA BOOLEANA

9. Apresente a tabela verdade a partir da expressão booleana encontrada.

R:

A	B	C	D	$A+B$	$C+D$	$(A+B) \cdot (C+D)$
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	0	0
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

12. Desenhe o circuito lógico que implementa a expressão $x = A\bar{B} + B\bar{C}(A+D)$. Após supor $A=0$, $B=0$, $C=1$ e $D=1$, determine pelo circuito lógico o valor de x .



$$0 \cdot \bar{0} + 0 \cdot \bar{1} \cdot (\bar{0} + 1) = x$$

$$0 \cdot 1 + 0 \cdot 0 \cdot (\bar{0}) = x$$

$$0 + 0 \cdot 0 = x$$

$$0 \cdot 0 = x$$

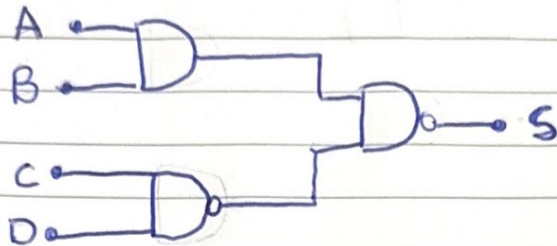
$$0 = x$$

14. Apresente a tabela Verdade A partir da expressão booleana. $S = AB + \bar{C} + CD$

A	B	C	D	AB	$\bar{C}$	CD	$AB + \bar{C} + CD$
0	0	0	0	0	1	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	0	0	0
0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	1
0	1	0	1	0	1	0	1
0	1	1	0	0	0	0	0
0	1	1	1	0	0	1	1
1	0	0	0	0	1	0	1
1	0	0	1	0	1	0	1
1	0	1	0	0	0	0	0
1	0	1	1	0	0	1	1
1	1	0	0	1	1	0	1
1	1	0	1	1	1	0	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	1	1

18. implemente o circuito lógico a partir da expressão booleana $S = \overline{AB(CD)}$

R:



21. Determine a tabela-verdade e a expressão booleana gerada pelo circuito.

R:

$$S = (\overline{A} \cdot B) \cdot (\overline{B+C}) \cdot (\overline{B+D})$$

A	B	C	D	$\overline{A} \cdot B$	$\overline{(B+C)}$	$\overline{(B+D)}$	S
0	0	0	0	0	1	1	0
0	0	0	1	0	1	0	0
0	0	1	0	0	0	1	0
0	0	1	1	0	0	0	0
0	1	0	0	1	0	0	0
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	0
0	1	1	1	1	0	0	0
1	0	0	0	0	1	1	0
1	0	0	1	0	1	0	0
1	0	1	0	0	0	1	0
1	0	1	1	0	0	0	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0

26. Determine a saída da porta XNOR.

R:

